

Digital strategies for the social sciences – 8

Geometric Data Analysis (2)

TODAY'S COURSE

Questions?

1st Project

2nd Project: June 8th 2025?

TODAY'S COURSE

Questions?

0. Review

1. GDA on categorical data: MCA (Multiple Correspondence Analysis)

2. GDA in practice

3. To go further

4. Case studies

REVIEW

An inductive approach

- Benzecri: “The model should follow the data, not the other way around”
- Assumption: Existence of a lower level embedding (= all dimensions / variables) are not necessary.
- Premised on the creation of new “factors”/“axes”

==> GDA as a linear / combination of the / original variables.

Goal: to extract relevant patterns while preserving as much information

REVIEW

An inductive approach

A configurational approach

- Not an attempt to “control” for other variables
(Reasoning “All things being unequal”)
- Trying to find the structure of a space
 - Bourdieu: Link between field theory & GDA

REVIEW

An inductive approach

A configurational approach

The role of interpretation

- Once the variables are selected, the researcher must interpret the axes.
- It comes after the analysis (good? bad?)
- This is done by looking
 - at the contribution of a variable to an axis
 - At the coordinates of a variable on an axis
 - (at its $\cos^2$, and other measures the quality of representation)

Introducing a data set

CASE STUDY

Valkompassen, 2014

- An investigation of the beliefs of voters, of politicians
- Carried out by SVT, every 4 years, with the help of political scientists
 - 45 questions on current topics, curated by political scientists
 - A series of socio-demographic information
 - In our case: MPs who responded (~245)

CASE STUDY

Example: Valkompassen, 2014

- An investigation of the beliefs of voters, of politicians

Questions:

- Can we sort candidates according on different value scales
- Is left/right enough to characterize SWE politics?
- How many dimensions structure Swedish politics?
- How cohesive are party in terms of values
- ...

CASE STUDY

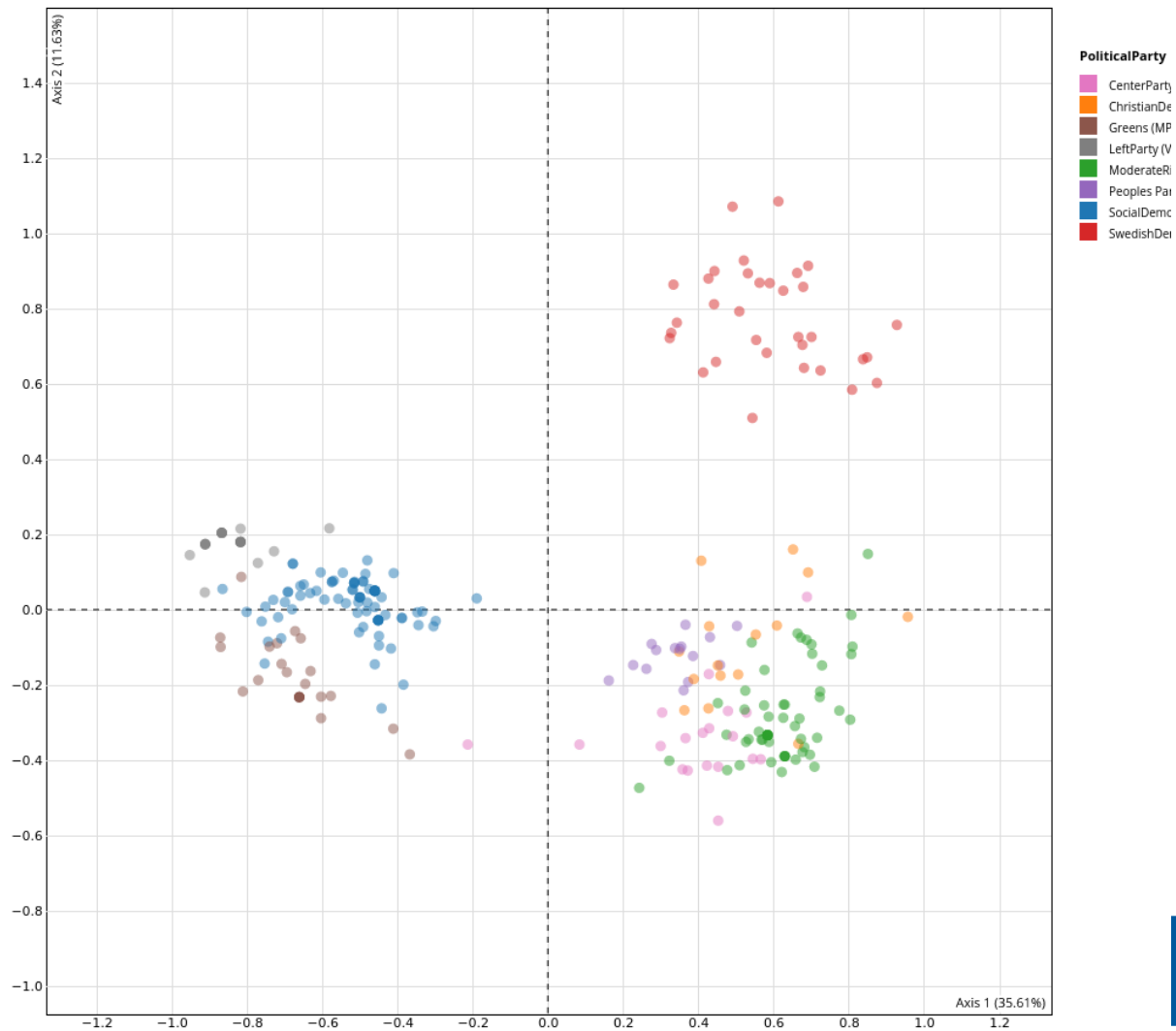
Example: Valkompassen, 2014

How to do this? GDA!

- Select your variables (why those?)
- Run your GDA
- Interpret it
- Comment

CASE STUDY

Example: Valkompassen, 2014



Geometric Data Analysis

PRINCIPLES OF MCA

Geometric Data Analysis

PRINCIPLES OF MCA

Often in the social sciences, the input variables are not continuous but categorical. This is often the case with responses to survey data.

Geometric Data Analysis

PRINCIPLES OF MCA

Example: A survey on leisure activities in France, 2000s

```
> head(cult)
```

	Leisure_Who	FreeTimeExperience	MoreTime	GoingOut	TV_week	Books_year	Gender	Diploma
1	Leisure:friends	lack time	home DIY	GoingOut:friends	TV:[19;30h[	5-12 books	women	CAP-BEP
2	Leisure:family	lack time	artistic activities	GoingOut:alone	TV:[10;19h[	5-12 books	men	BEPC
3	Leisure:family	lack time	home DIY	GoingOut:partner	TV:[19;30h[	1-4 books	men	>Bac+2
4	Leisure:family	lack time	to rest	GoingOut:partner	TV:[10;19h[	no book	men	Bac+2
5	Leisure:alone	lack time	develop knowledge	don't go out	TV:>=30h	1-4 books	women	no degree
6	Leisure:friends	always sth to do	physical activities	don't go out	TV:>=30h	no book	women	CEP

	AgeGroup	PCS
1	35-45	Housewives
2	>65	Retired
3	<=25	Students
4	<=25	No occupation -others
5	45-55	Housewives
6	35-45	Housewives

6 variables about practices, 4 about socio-demographic
circa 2400 respondents

Geometric Data Analysis

PRINCIPLES OF MCA

Often in the social sciences, the input variables are not continuous but categorical. This is often the case with responses to survey data.

To do GDA on categorical data, one does **MCA** (Multiple correspondence analysis)

The intuition is the same, the maths vary a bit.

The interpretation remains the same

Geometric Data Analysis

PRINCIPLES OF MCA

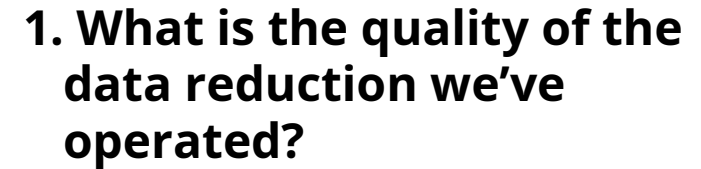
MCA is used on categorical variables

From PCA to MCA:

- Create a 'complete disjunctive table' (CDT)
- Apply PCA

Question: What will change?

MCA: HOW TO READ IT



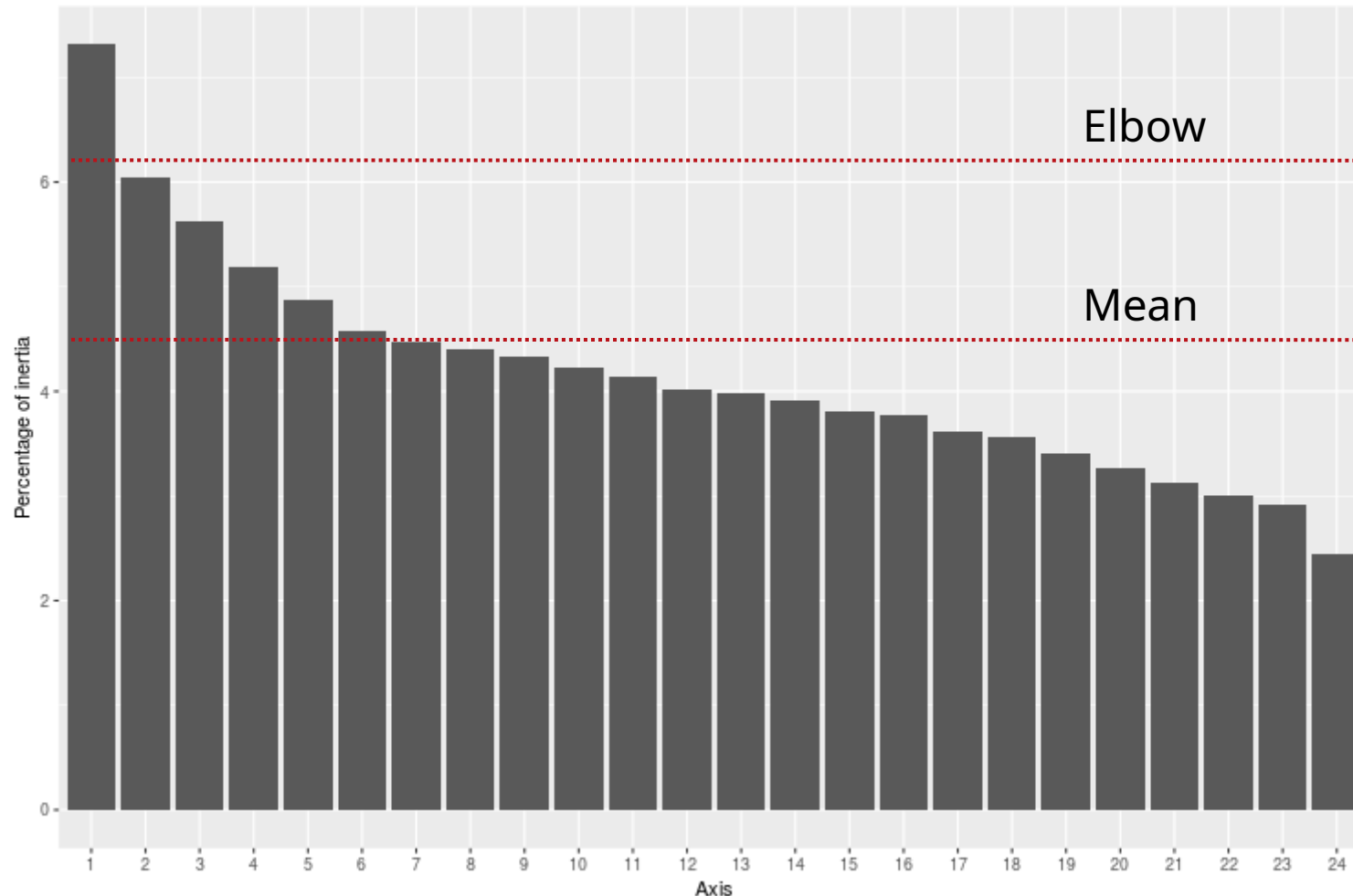
- Measured by the eigenvalue (the variance of the axis)
- Here: $7.3 + 6.02$
- ~ 25 levels (6 variables, 4 or 5 levels each)

Geometric Data Analysis

MCA: HOW TO READ IT

1b. How many axes to retain?

Eigenvalues histogram



Several criteria

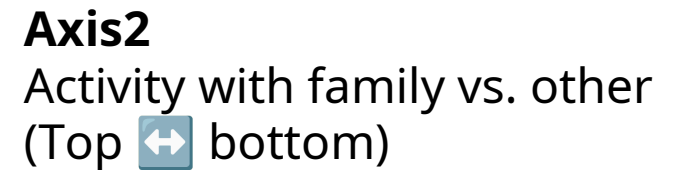
- Elbow
- Mean
- Interpretability

Keep in mind: what is your goal ?

2. Which variables are opposed on each axis?



3. Provide an interpretation for each axis



MCA: HOW TO READ IT

3. Analysis the axes

Contribution to the axis

A central information to interpret the axes

Denotes the share of each variable in the creation of this new axis.

The higher, the more important in the production of the new axis.

Dimension		Active variables				
Axis 1 (58.39%)		Show 10 entries		Search:		
Variable	Coord	Contrib	Cos2	Cor		
General_Radio	0.896	19.65	0.803	0.896		
X24_7_NewsNetwork	0.848	17.57	0.718	0.848		
NewRadio	0.847	17.55	0.717	0.847		
CSPAN_Senate	0.835	17.04	0.696	0.835		
LocalRadio	0.775	14.69	0.600	0.775		
CSPAN	0.545	7.27	0.297	0.545		
Tv	0.505	6.23	0.255	0.505		
Showing 1 to 7 of 7 entries					Previous	Next

MCA: HOW TO READ IT

3. Analysis the axes

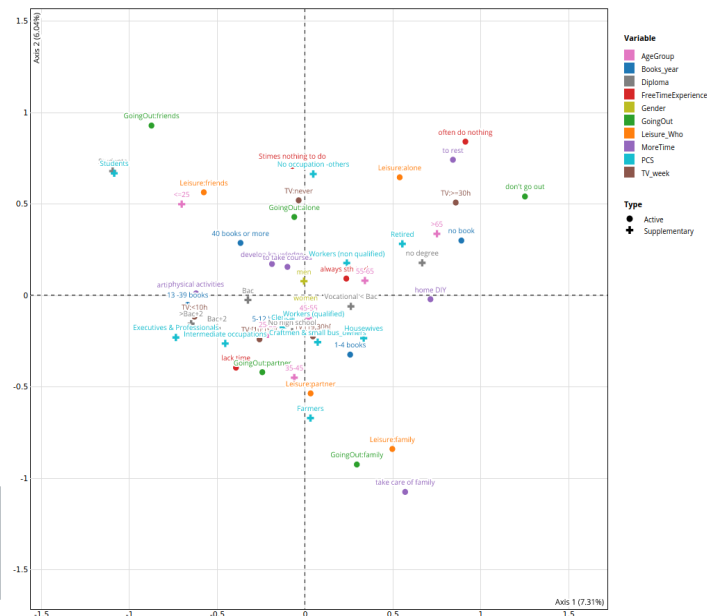
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MCA: HOW TO READ IT

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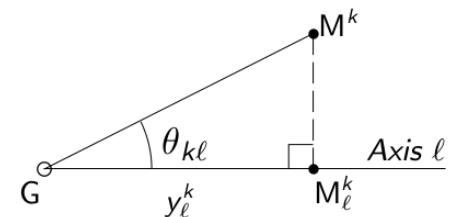
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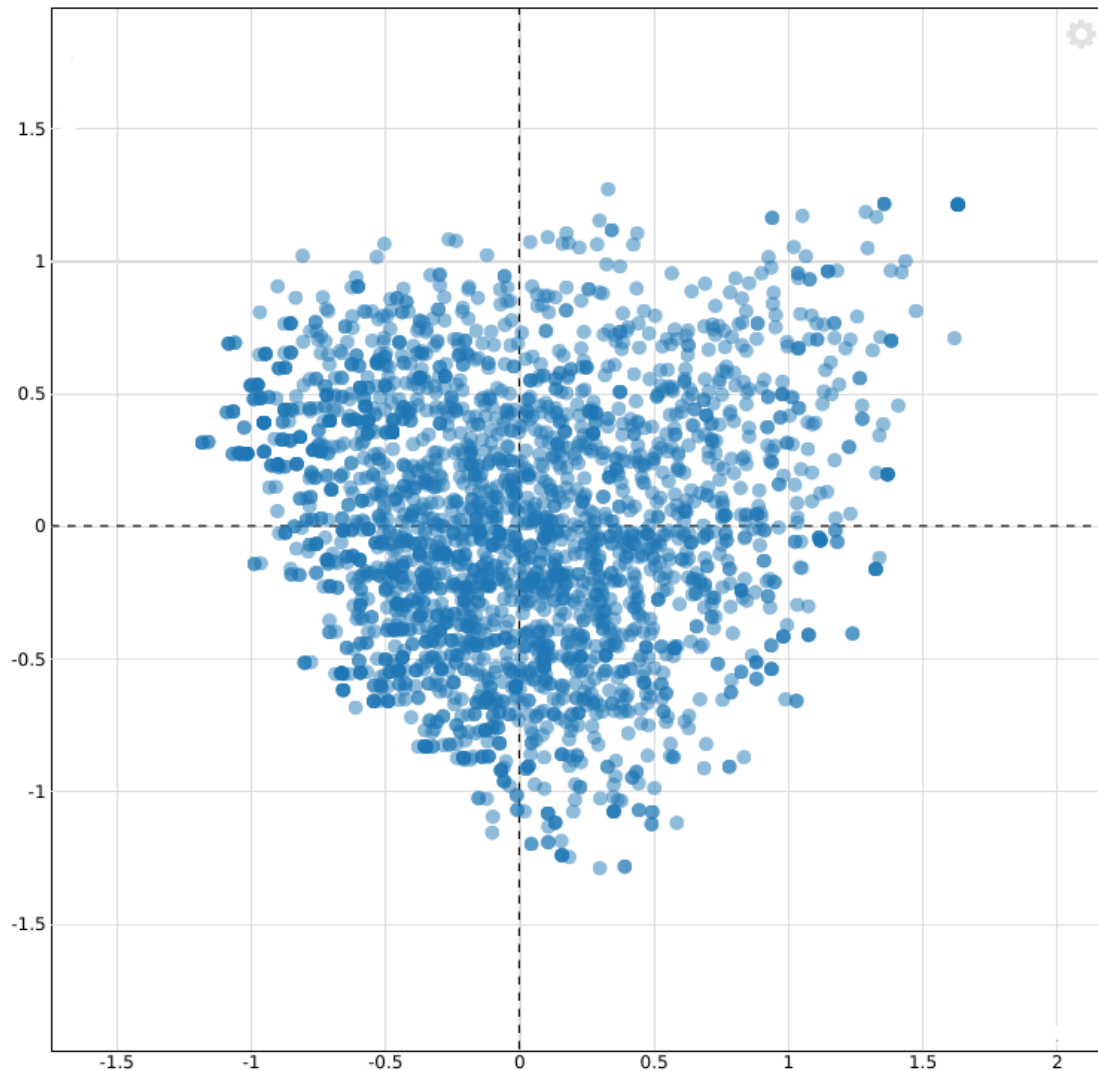
Square cosine: Denotes the quality of the representation of a given point on a given axis.

Range is between 0 and 1. The higher, the better.



Geometric Data Analysis

MCA: HOW TO READ IT

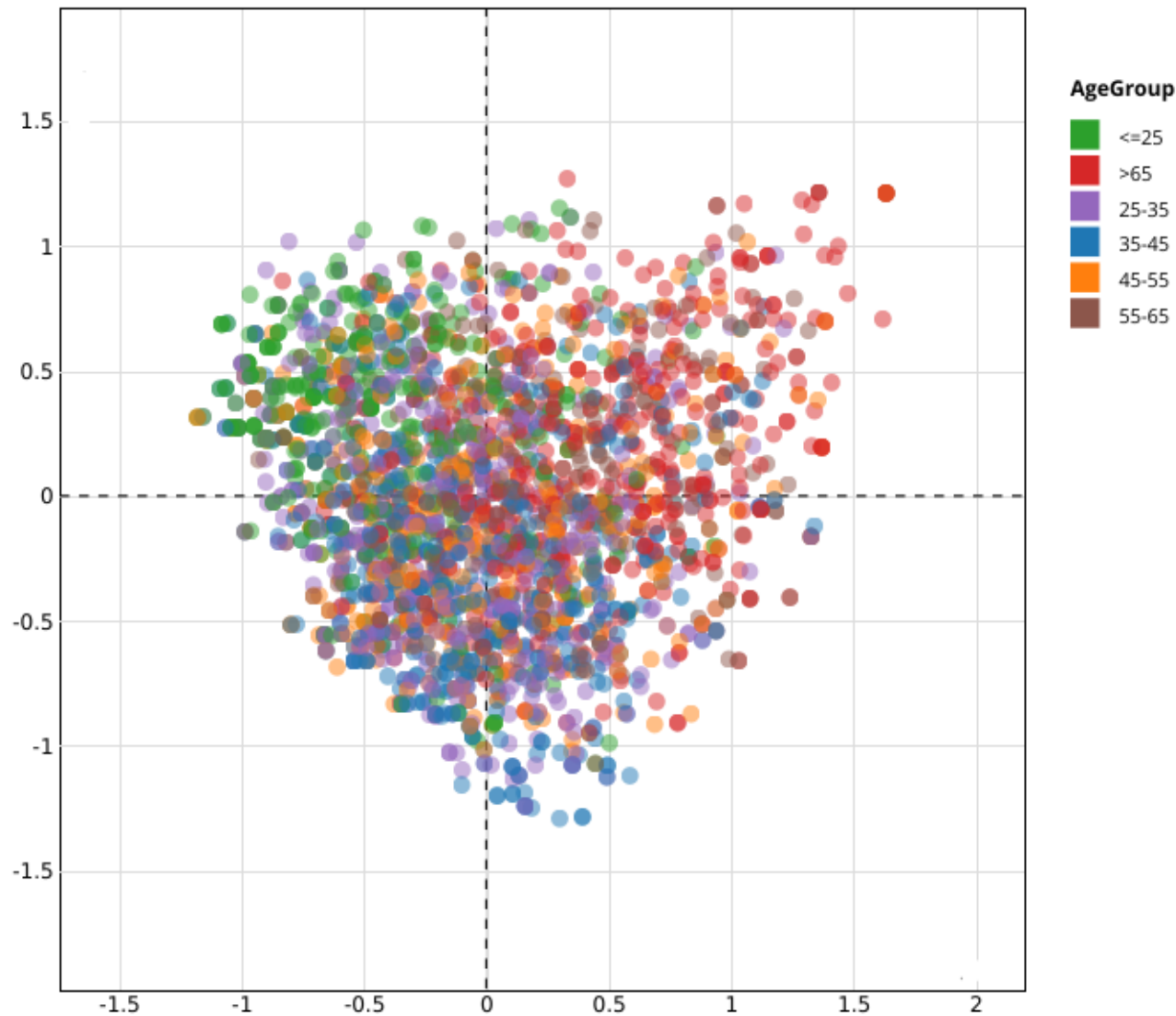


4. You can then look at where the individuals stand

Geometric Data Analysis

MCA: HOW TO READ IT

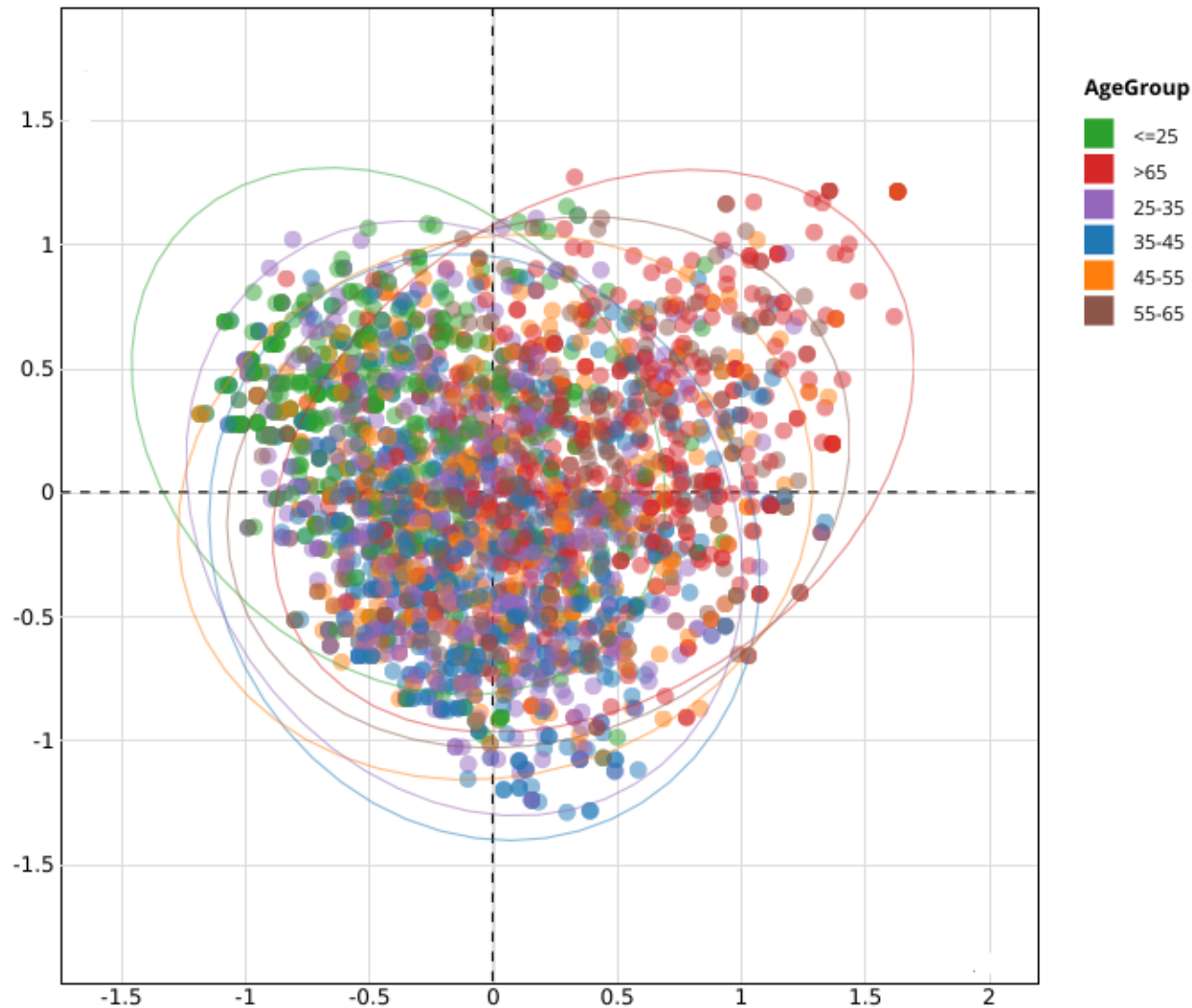
5. You can color plot by supplementary variable



Geometric Data Analysis

MCA: HOW TO READ IT

5. You can color plot by supplementary variable



Beyond eyesight:
Confidence ellipses

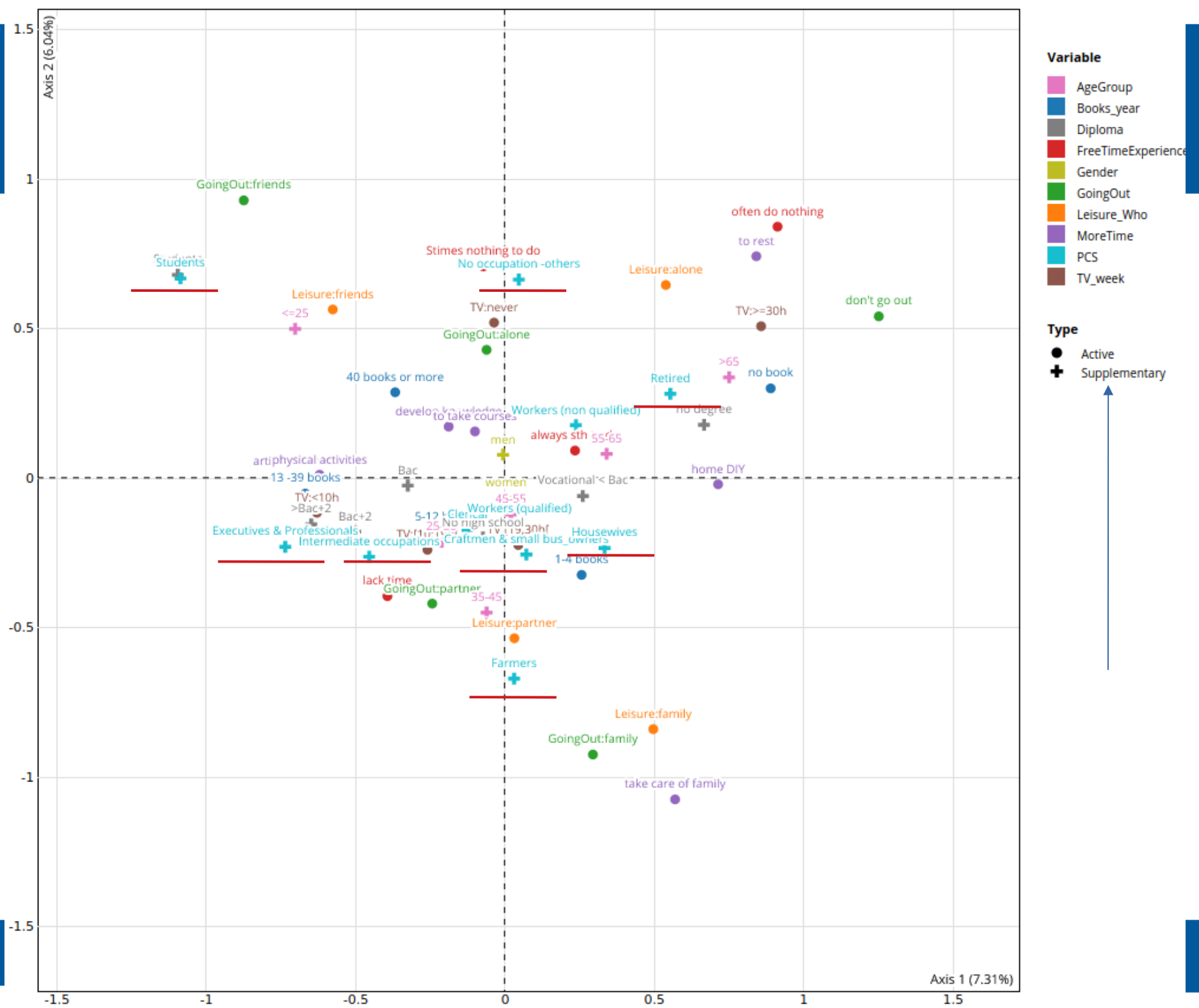
Geometric Data Analysis

MCA: HOW TO READ IT

6. Or you can *project these* 'supplementary variables'

Supplementary variables do not enter into the calculation of the axes
But they can be projected onto the 2D plan.

To do so, the barycenter of these variables is calculated and displayed



Geometric Data Analysis

MCA: HOW TO READ IT

6. Or you can project 'supplementary variables'

Supplementary variables do not enter into the calculation of the axes
But they can be projected onto the 2D plan.

To do so, the barycenter of these variables is calculated and displayed

This is useful to interpret, to see more correlations and to move towards some explanatory purposes, BUT keep in mind that:

- The projected point point is just the average of all coordinates
 - These are mere POSSIBLE correlations
- USE WITH CAUTION

MCA IN PRACTICE

Data preparation

Depends on the type of technique. For MCA

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1. Avoid hugely unbalanced data sets. At any rate, avoid having categories of a variable with less than 5% of the total population, avoid having over 60% of one category. [relaxed in machine learning]

MCA IN PRACTICE

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2. Avoid NAs (unless they are meaningful and satisfy 1), as they tend to “twist” the axes in a non-relevant way.

Trick: if too many NAs in a variable, consider dropping it, recoding it, or place it as a supplementary variable.

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Trick: if too many NAs in a variable, consider dropping it, recoding it, or place it as a supplementary variable (to be defined soon).

3. Unless you do a really exploratory analysis, avoid putting “all the variables at hand” in the MCA, as they all carry the same weight.

On this note: avoid having very unbalanced set of variables (say 80% dealing with the economy, and 20% with social and political issues)

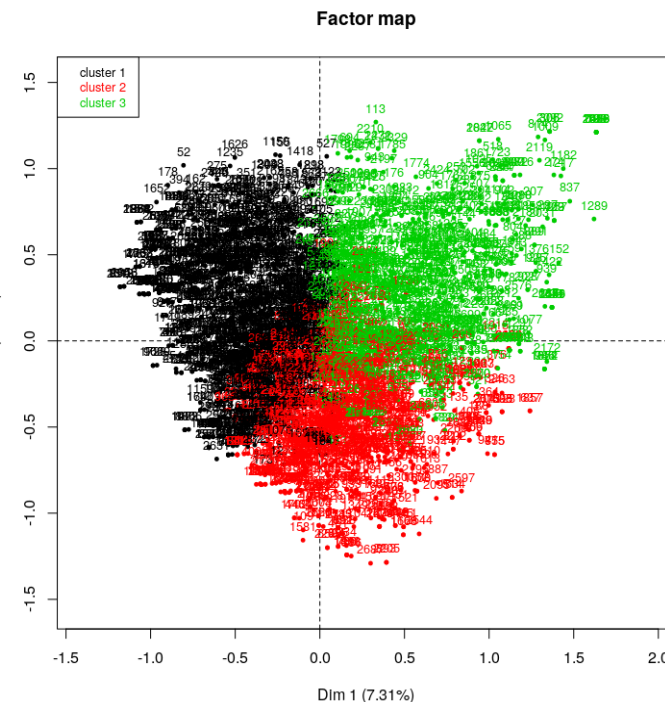
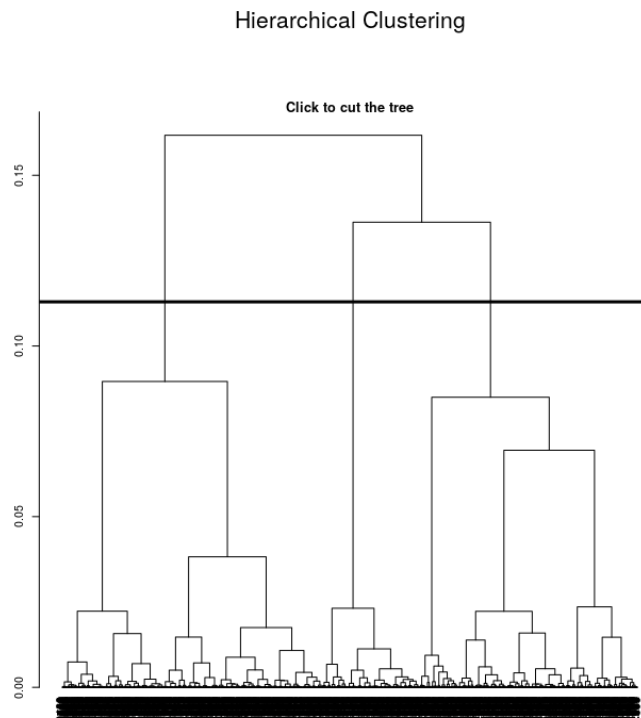
TO GO FURTHER

There are many developments, and many ways to go further.
For the purpose of this course, I am just pointing to a few classic ones

1. Classification

One may want to capture groups of individuals. To do so, a common way is to do a classification on the result of the analysis.

In R, **HCPC(res)** will perform a hierarchical clustering



TO GO FURTHER

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2. Specific MCA

A technique meant to deal with NAs. Developed in the 2000s.

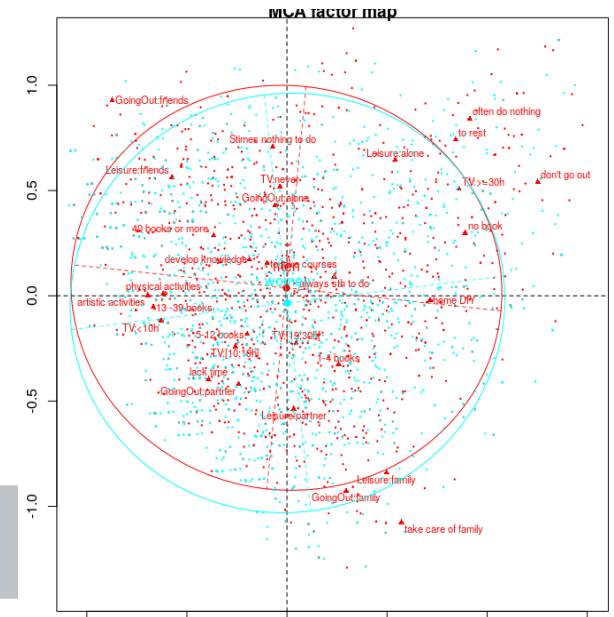
3. Class specific MCA

A technique meant to focus on a small group (for instance: Oxbridge in UK population), while keeping them positioned within the global social space.

4. Concentration Ellipses

A technique meant to visualize differences with respect to the supplementary variables. Takes the shape of a circle (per level) applied to the cloud of points.

Here: men and women on the culture example



TO GO FURTHER

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Two libraries come in handy to do these things

1. **factoextra**, the sequel to FactoMiner, for refined visual analysis
2. **GDAtools** for doing all the speMCA and CS analyzes.

Check them out on CRAN

<https://cran.r-project.org/web/packages/GDAtools/GDAtools.pdf>